

Minor Project 2: Vet Clinic Manager

Windows Desktop Application (WPF + SQL Server)

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Repository: <https://github.com/adnvanv/547> (folder minorProj2)

Overview

Vet Clinic Manager is a WPF desktop application for a small veterinary clinic. Staff can manage pet owners, their pets and the pets' appointments, and an administrator can manage the clinic's veterinarian records. Data is stored in a new SQL Server database, VetClinicDB, and every SQL statement lives in a separate data-layer project.

Solution structure

- VetClinic.Data (class library, net10.0): models plus one repository per table (OwnerRepository, PetRepository, AppointmentRepository, VeterinarianRepository) built on ADO.NET with Microsoft.Data.SqlClient. All values are passed as SQL parameters.
- VetClinic.App (WPF, net10.0-windows): the user interface. It references VetClinic.Data and never builds SQL itself.
- Scripts: 01_VetClinicDB_DDL.sql creates the database and tables; 02_VetClinicDB_DML.sql seeds sample data.

Database (4 tables)

Table	Purpose	Relationships
Owners	Clients: name, email, phone, city	Parent of Pets
Pets	Name, species, breed, birth date, weight	FK to Owners (ON DELETE CASCADE)
Appointments	Date/time, reason, status, cost	FK to Pets (CASCADE) and Veterinarians
Veterinarians	Staff records (admin-only)	Referenced by Appointments

How requirements are met

Requirement	Where it is implemented
Windows desktop app	WPF (.NET 10) with the built-in Fluent theme
New database, 3+ tables	VetClinicDB with Owners, Pets, Appointments, Veterinarians

Requirement	Where it is implemented
Separate data layer	VetClinic.Data class library project in the same solution
Full CRUD on 3 tables	Owners, Pets and Appointments tabs (plus Veterinarians on the Administration tab)
Search on 2+ text fields	Owners tab: search by ID, first name, last name, email, phone, city, or any field, using SQL LIKE '%term%'
Settings page	Theme (Windows/Light/Dark), text size, delete confirmation, administrator mode, connection string; saved to %APPDATA%\VetClinic\settings.json
Administration page	Veterinarians can only be added, edited or deleted on the Administration tab, which is locked until administrator mode is turned on
No security	No login; administrator mode is a simple settings toggle

Running the project

1. Run VetClinic/Scripts/01_VetClinicDB_DDL.sql and then 02_VetClinicDB_DML.sql against SQL Server Express (.\SQLEXPRESS).
2. Open VetClinic/Code/VetClinic.sln in Visual Studio, or run: dotnet run --project VetClinic/Code/VetClinic.App
3. If your server name differs, change the connection string on the Settings tab and click Test connection.

Screenshots

All screenshots were taken of the running application on my computer against the local SQL Server Express database.

Search

The screenshot shows the 'Vet Clinic Manager' application window. The title bar reads 'Vet Clinic Manager'. Below the title bar, the subtitle is 'Owners, pets and appointments for the clinic'. A navigation bar contains links: 'Owners', 'Pets', 'Appointments', 'Administration', and 'Settings'. The 'Owners' section is active, displaying a search interface. A dropdown menu for 'Last name' is set to 'carter'. A 'Search' button is highlighted, and a 'Clear' button is next to it. Below the search bar, a table lists two owners matching the search criteria:

ID	First name	Last name	Email	Phone	City	Pets
1	Emily	Carter	emily.carter@example.com	(614) 555-0101	Columbus	2
6	Ethan	Carter	ethan.carter@example.com	(614) 555-0175	Westerville	1

Below the table, a status bar indicates '2 owner(s) matching "carter" in Last name' and 'Ready'. To the right of the table, the 'New owner' form is visible, with fields for ID (set to '(assigned on save)'), First name, Last name, Email, Phone, and City. Below these fields are 'New', 'Save', and 'Delete' buttons. A note at the bottom right states: 'Deleting an owner also deletes their pets and those pets' appointments.'

Figure 1. Search by last name: "carter" matches two owners.

Vet Clinic Manager

Owners, pets and appointments for the clinic

Owners Pets Appointments Administration Settings

Owners

City

columbus

Search

Clear

ID	First name	Last name	Email	Phone	City	Pets
1	Emily	Carter	emily.carter@example.com	(614) 555-0101	Columbus	2
4	Michael	Johnson	mjohnson@example.com	(614) 555-0199	Columbus	1

2 owner(s) matching "columbus" in City

Ready

New owner

ID (assigned on save)

First name *

Last name *

Email

Phone

City

New

Save

Delete

Deleting an owner also deletes their pets and those pets' appointments.

Figure 2. Search by city: "columbus" matches the Columbus owners.

Vet Clinic Manager

Owners, pets and appointments for the clinic

Owners Pets Appointments Administration Settings

Owners

Email

brown@

Search

Clear

ID	First name	Last name	Email	Phone	City	Pets
5	Olivia	Brown	olivia.brown@example.com	(937) 555-0113	Dayton	1

1 owner(s) matching "brown@" in Email

Ready

New owner

ID (assigned on save)

First name *

Last name *

Email

Phone

City

New

Save

Delete

Deleting an owner also deletes their pets and those pets' appointments.

Figure 3. Search by email: "brown@" finds Olivia Brown.

Insert

The screenshot shows the 'Vet Clinic Manager' application with the 'Owners' tab selected. On the left, a table lists 8 existing owners. On the right, the 'New owner' form is open, with fields for ID (assigned on save), First name (Grace), Last name (Hopper), Email (grace.hopper@example.com), Phone ((614) 555-0177), and City (Columbus). A 'Save' button is highlighted in red.

Vet Clinic Manager
Owners, pets and appointments for the clinic

Owners Pets Appointments Administration Settings

Owners

Any field Search Clear

ID	First name	Last name	Email	Phone	City	Pets
5	Olivia	Brown	olivia.brown@example.com	(937) 555-0113	Dayton	1
1	Emily	Carter	emily.carter@example.com	(614) 555-0101	Columbus	2
6	Ethan	Carter	ethan.carter@example.com	(614) 555-0175	Westerville	1
4	Michael	Johnson	mjohnson@example.com	(614) 555-0199	Columbus	1
3	Sophia	Martinez	sophia.m@example.com	(740) 555-0188	Delaware	1
2	Daniel	Nguyen	dnguyen@example.com	(614) 555-0142	Dublin	1
8	Noah	Patel	noah.patel@example.com	(614) 555-0120	Hilliard	2
7	Ava	Wilson	ava.wilson@example.com	(513) 555-0164	Cincinnati	1

8 owner(s)
Ready

New owner

ID (assigned on save)

First name * Grace

Last name * Hopper

Email grace.hopper@example.com

Phone (614) 555-0177

City Columbus

New Save Delete

Deleting an owner also deletes their pets and those pets' appointments.

Figure 4. Insert: filling in a new owner (Grace Hopper) after clicking New.

The screenshot shows the 'Vet Clinic Manager' application with the 'Owners' tab selected. The table now lists 9 owners, with the new entry 'Grace Hopper' highlighted in orange. On the right, the 'Edit owner #9' form is open, showing the same details as in Figure 4. A 'Save' button is highlighted in red.

Vet Clinic Manager
Owners, pets and appointments for the clinic

Owners Pets Appointments Administration Settings

Owners

Any field Search Clear

ID	First name	Last name	Email	Phone	City	Pets
5	Olivia	Brown	olivia.brown@example.com	(937) 555-0113	Dayton	1
1	Emily	Carter	emily.carter@example.com	(614) 555-0101	Columbus	2
6	Ethan	Carter	ethan.carter@example.com	(614) 555-0175	Westerville	1
9	Grace	Hopper	grace.hopper@example.com	(614) 555-0177	Columbus	0
4	Michael	Johnson	mjohnson@example.com	(614) 555-0199	Columbus	1
3	Sophia	Martinez	sophia.m@example.com	(740) 555-0188	Delaware	1
2	Daniel	Nguyen	dnguyen@example.com	(614) 555-0142	Dublin	1
8	Noah	Patel	noah.patel@example.com	(614) 555-0120	Hilliard	2
7	Ava	Wilson	ava.wilson@example.com	(513) 555-0164	Cincinnati	1

9 owner(s)
3:44 PM Added owner #9 Grace Hopper.

Edit owner #9

ID 9

First name * Grace

Last name * Hopper

Email grace.hopper@example.com

Phone (614) 555-0177

City Columbus

New Save Delete

Deleting an owner also deletes their pets and those pets' appointments.

Figure 5. Insert: after Save the owner gets ID 9 and appears in the list.

Vet Clinic Manager

Owners, pets and appointments for the clinic

Owners

Pets

Appointments

Administration

Settings

Pets

ID	Name	Species	Breed	Owner	Born	Weight (kg)
1	Biscuit	Dog	Golden Retriever	Emily Carter	4/12/2019	31.50
10	Daisy	Dog	Corgi	Noah Patel	12/24/2020	12.40
5	Kiwi	Bird	Cockatiel	Michael Johnson	6/1/2022	0.09
4	Luna	Cat	Maine Coon	Sophia Martinez	2/14/2020	6.80
7	Max	Dog	Beagle	Ethan Carter	9/9/2018	11.30
2	Mochi	Cat	Siamese	Emily Carter	8/3/2021	4.20
9	Nugget	Hamster	Syrian	Noah Patel	3/10/2024	0.15
6	Pepper	Rabbit	Holland Lop	Olivia Brown	1/25/2023	1.60
3	Rex	Dog	German Shepherd	Daniel Nguyen	11/20/2017	38.00
8	Shadow	Cat	Domestic Shorthair	Ava Wilson	5/30/2016	5.10

10 pet(s)

3:44 PM Deleted owner #9 Grace Hopper.

New pet

ID

(assigned on save)

Owner *

Patel, Noah

Name *

Pixel

Species *

Cat

Breed

Bengal

Birth date

Select a date

Weight (kg)

4.8

New

Save

Delete

Deleting a pet also deletes its appointments.

Figure 6. Insert on the Pets table: a new cat, Pixel, for owner Noah Patel.

Vet Clinic Manager

Owners, pets and appointments for the clinic

Owners

Pets

Appointments

Administration

Settings

Administration: Veterinarians

ID	First name	Last name	Specialty	Active	Appointments
1	Maria	Alvarez	General Practice	☒	4
4	Thomas	Becker	Exotic Animals	☐	1
5	Hannah	Kim	Cardiology	☒	0
2	James	Okafor	Surgery	☒	2
3	Priya	Raman	Dermatology	☒	2

5 veterinarian(s), 4 active

3:46 PM Added veterinarian #5.

Edit veterinarian #5

ID

5

First name *

Hannah

Last name *

Kim

Specialty

Cardiology

Status

☒ Active (can be booked)

New

Save

Delete

A veterinarian with appointments on record cannot be deleted; mark them inactive instead.

Figure 7. Insert on the admin-only Veterinarians table.

Update

Vet Clinic Manager

Owners, pets and appointments for the clinic

Owners Pets Appointments Administration Settings

Owners

Any field

Search

Clear

ID	First name	Last name	Email	Phone	City	Pets
5	Olivia	Brown	olivia.brown@example.com	(937) 555-0113	Dayton	1
1	Emily	Carter	emily.carter@example.com	(614) 555-0101	Columbus	2
6	Ethan	Carter	ethan.carter@example.com	(614) 555-0175	Westerville	1
9	Grace	Hopper	grace.hopper@example.com	(614) 555-0177	Columbus	0
4	Michael	Johnson	mjohnson@example.com	(614) 555-0199	Columbus	1
3	Sophia	Martinez	sophia.m@example.com	(740) 555-0188	Delaware	1
2	Daniel	Nguyen	dnguyen@example.com	(614) 555-0142	Dublin	1
8	Noah	Patel	noah.patel@example.com	(614) 555-0120	Hilliard	2
7	Ava	Wilson	ava.wilson@example.com	(513) 555-0164	Cincinnati	1

9 owner(s)

3:44 PM Added owner #9 Grace Hopper.

Edit owner #9

ID

9

First name *

Grace

Last name *

Hopper

Email

grace.hopper@example.com

Phone

(614) 555-0200

City

Worthington

New

Save

Delete

Deleting an owner also deletes their pets and those pets' appointments.

Figure 8. Update: changing Grace Hopper's phone number and city.

Vet Clinic Manager

Owners, pets and appointments for the clinic

Owners Pets Appointments Administration Settings

Owners

Any field

Search

Clear

ID	First name	Last name	Email	Phone	City	Pets
5	Olivia	Brown	olivia.brown@example.com	(937) 555-0113	Dayton	1
1	Emily	Carter	emily.carter@example.com	(614) 555-0101	Columbus	2
6	Ethan	Carter	ethan.carter@example.com	(614) 555-0175	Westerville	1
9	Grace	Hopper	grace.hopper@example.com	(614) 555-0177	Columbus	0
4	Michael	Johnson	mjohnson@example.com	(614) 555-0199	Columbus	1
3	Sophia	Martinez	sophia.m@example.com	(740) 555-0188	Delaware	1
2	Daniel	Nguyen	dnguyen@example.com	(614) 555-0142	Dublin	1
8	Noah	Patel	noah.patel@example.com	(614) 555-0120	Hilliard	2
7	Ava	Wilson	ava.wilson@example.com	(513) 555-0164	Cincinnati	1

9 owner(s)

3:44 PM Added owner #9 Grace Hopper.

Edit owner #9

ID

9

First name *

Grace

Last name *

Hopper

Email

grace.hopper@example.com

Phone

(614) 555-0200

City

Worthington

New

Save

Delete

Deleting an owner also deletes their pets and those pets' appointments.

Figure 9. Update: the grid and status bar show the saved changes.

Vet Clinic Manager

Owners, pets and appointments for the clinic

Owners Pets Appointments Administration Settings

Appointments

ID	When	Pet	Owner	Veterinarian	Reason	Status	Cost
8	10/5/2026 8:45 AM	Daisy	Noah Patel	Dr. James Okafor	Spay surgery	Scheduled	
7	10/1/2026 2:00 PM	Shadow	Ava Wilson	Dr. Priya Raman	Allergy follow-up	Scheduled	
6	9/28/2026 9:30 AM	Max	Ethan Carter	Dr. Maria Alvarez	Dental cleaning	Completed	\$180.00
4	9/18/2026 3:00 PM	Luna	Sophia Martinez	Dr. Maria Alvarez	Vaccinations	Completed	\$65.00
9	9/10/2026 4:00 PM	Pepper	Olivia Brown	Dr. Maria Alvarez	Check-up	Cancelled	
3	9/2/2026 10:15 AM	Rex	Daniel Nguyen	Dr. James Okafor	Hip X-rays	Completed	\$240.00
2	8/20/2026 1:30 PM	Mochi	Emily Carter	Dr. Priya Raman	Skin rash on neck	Completed	\$120.00
1	8/14/2026 9:00 AM	Biscuit	Emily Carter	Dr. Maria Alvarez	Annual wellness exam	Completed	\$85.00
5	7/11/2026 11:00 AM	Kiwi	Michael Johnson	Dr. Thomas Becker	Beak and nail trim	Completed	\$40.00

9 appointment(s)

3:45 PM Updated appointment #6.

Edit appointment #6

ID

6

Pet *

Max (Dog) - Ethan Carter

Veterinarian *

Dr. Maria Alvarez

Date *

9/28/2026

Time *

9:30 AM

Reason *

Dental cleaning

Status

Completed

Cost (\$)

180.00

New

Save

Delete

Only active veterinarians can be booked. Veterinarians are managed on the Administration tab.

Figure 10. Update on the Appointments table: marking appointment #6 Completed with a \$180 cost.

Vet Clinic Manager

Owners, pets and appointments for the clinic

Owners Pets Appointments Administration Settings

Administration: Veterinarians

ID	First name	Last name	Specialty	Active	Appointments
1	Maria	Alvarez	General Practice	☒	4
4	Thomas	Becker	Exotic Animals	☐	1
5	Hannah	Kim	Cardiology and Internal Medicine	☒	0
2	James	Okafor	Surgery	☒	2
3	Priya	Raman	Dermatology	☒	2

5 veterinarian(s), 4 active

3:46 PM Updated veterinarian #5.

Edit veterinarian #5

ID

5

First name *

Hannah

Last name *

Kim

Specialty

Cardiology and Internal Medicine

Status

☒ Active (can be booked)

New

Save

Delete

A veterinarian with appointments on record cannot be deleted; mark them inactive instead.

Figure 11. Update on the Veterinarians table (Administration tab).

Delete

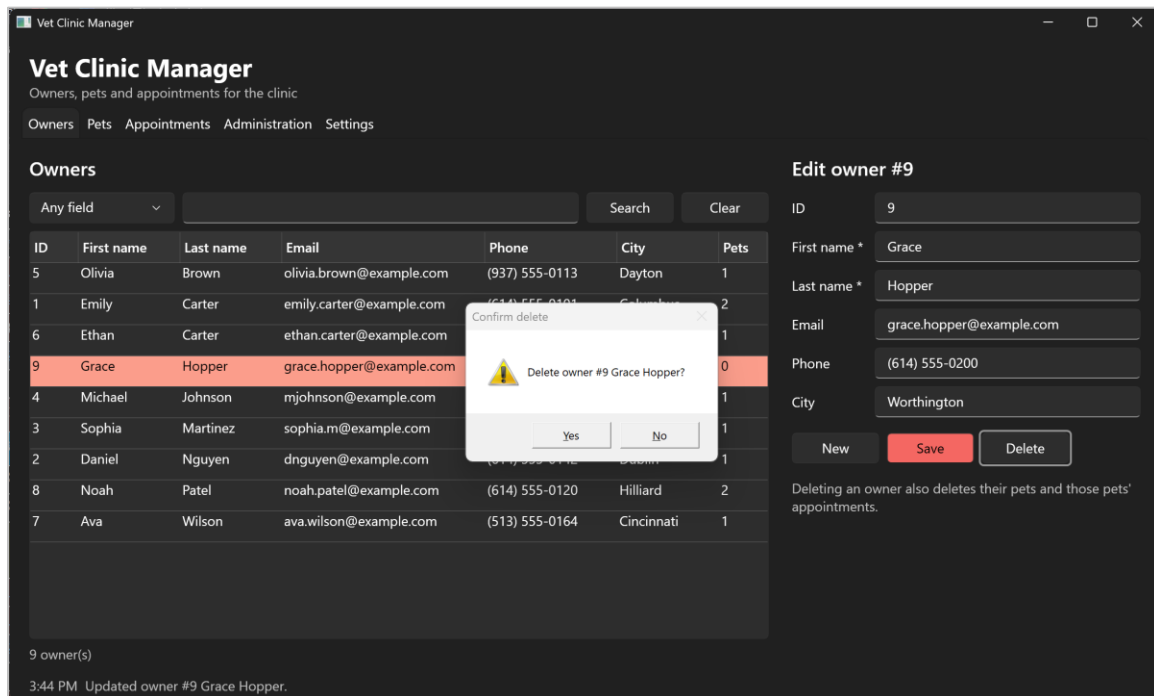


Figure 12. Delete: the confirmation prompt before deleting owner #9.

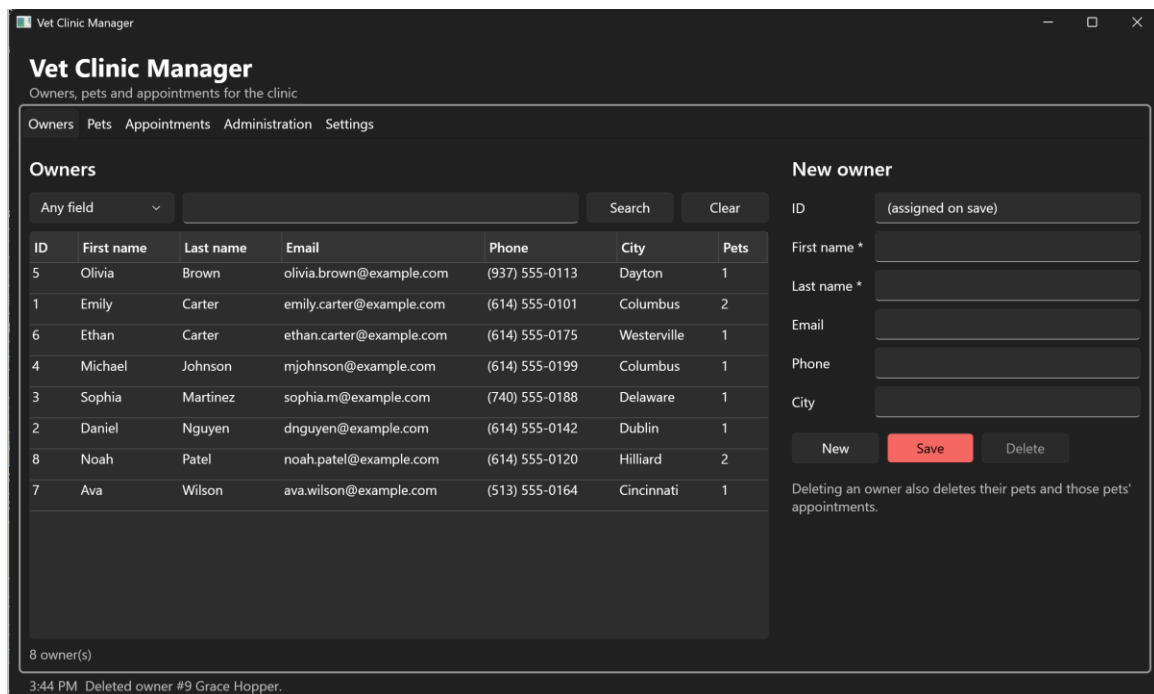


Figure 13. Delete: owner #9 is removed (8 owners remain).

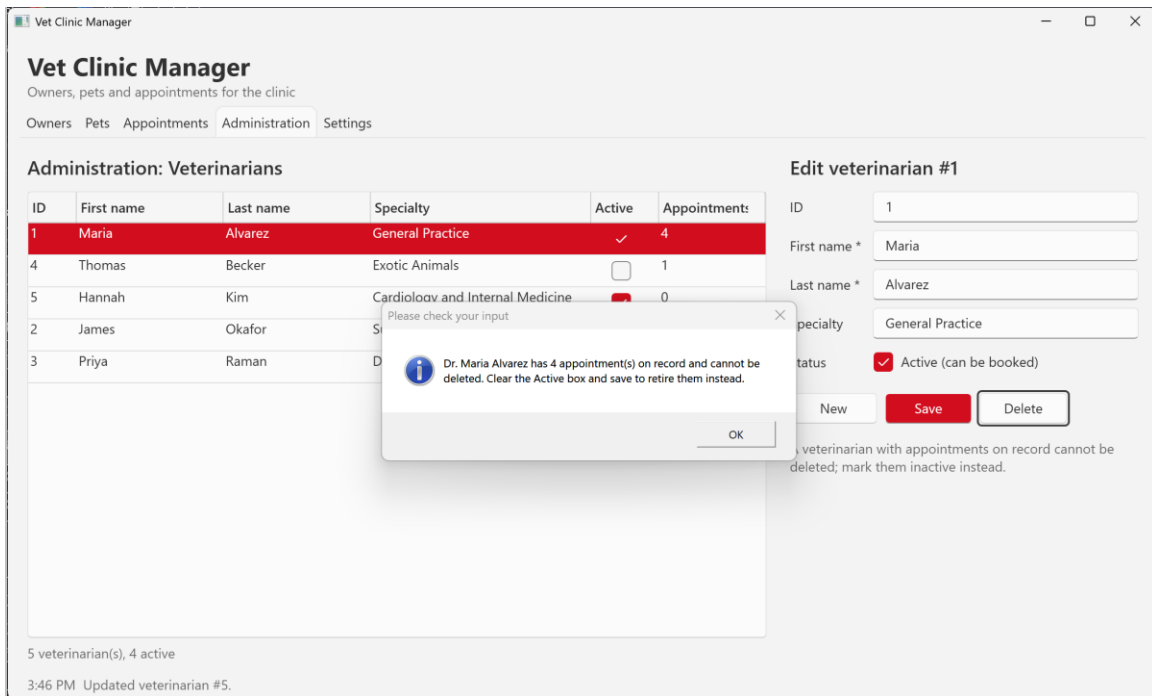


Figure 14. Delete protection: a veterinarian with appointments cannot be deleted and should be marked inactive instead.

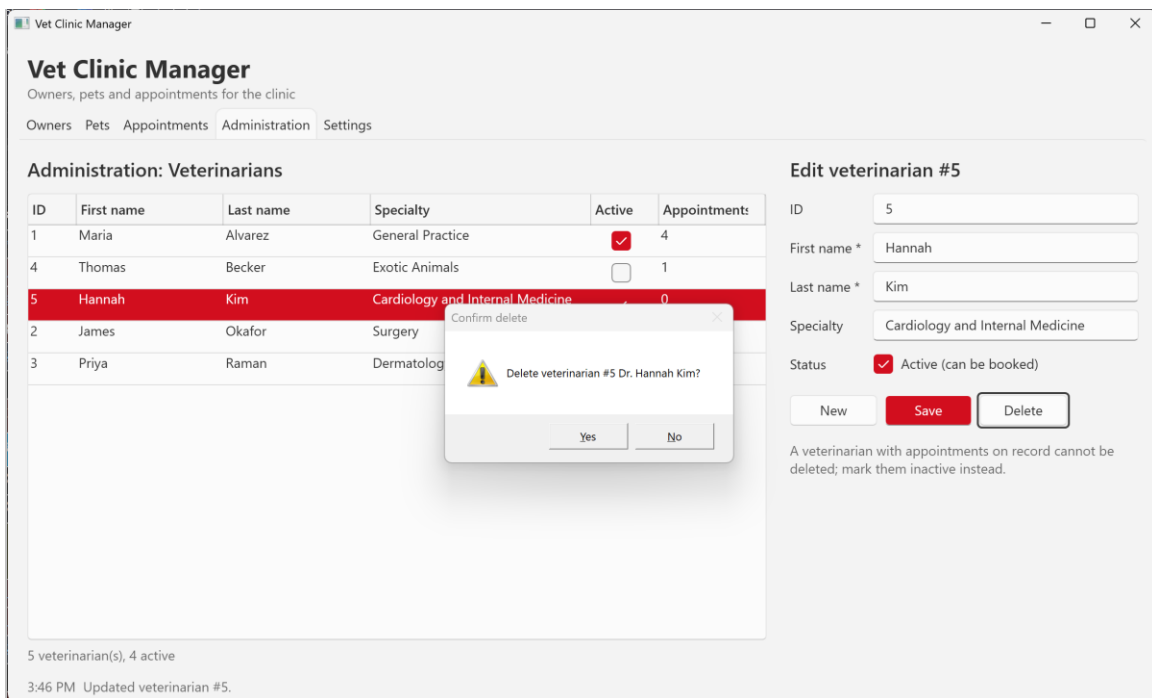


Figure 15. Delete on the Veterinarians table: confirming removal of the new veterinarian.

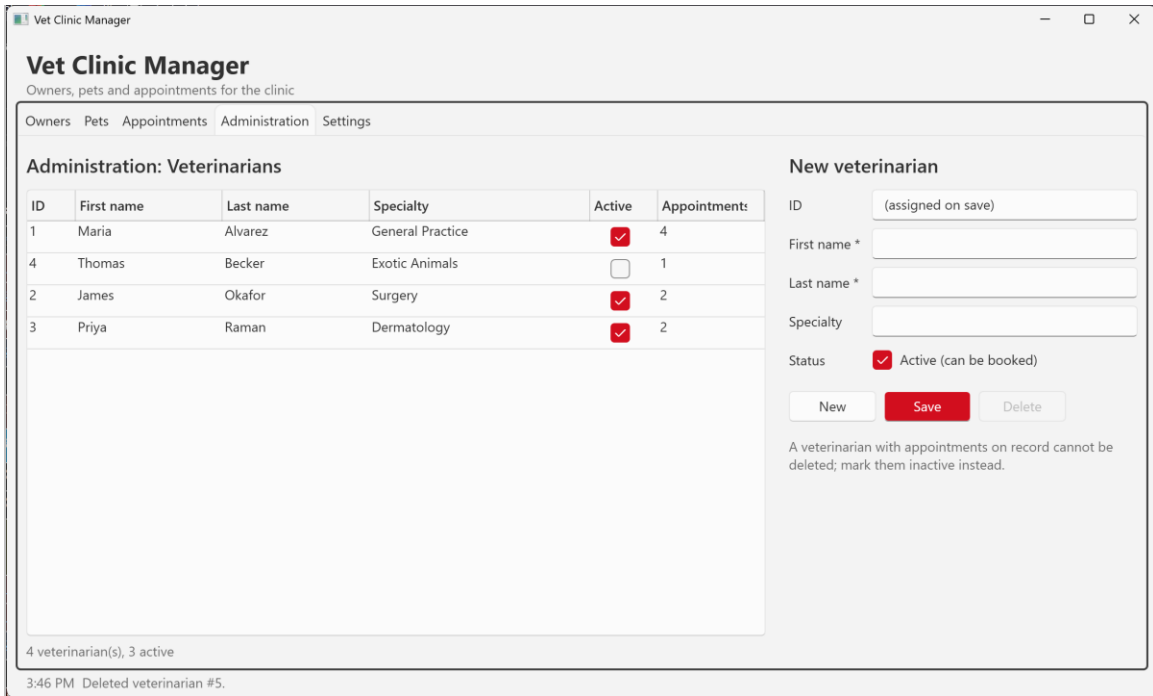


Figure 16. Delete: the veterinarian is removed.

Settings and Administration

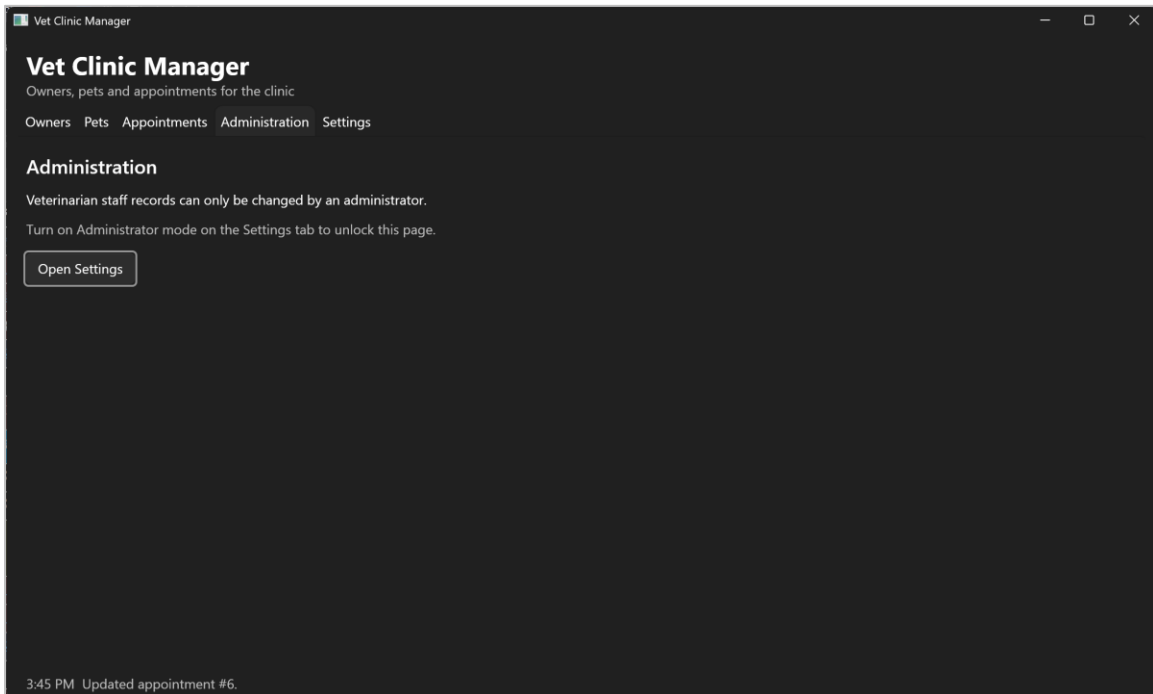


Figure 17. The Administration tab is locked until administrator mode is enabled.

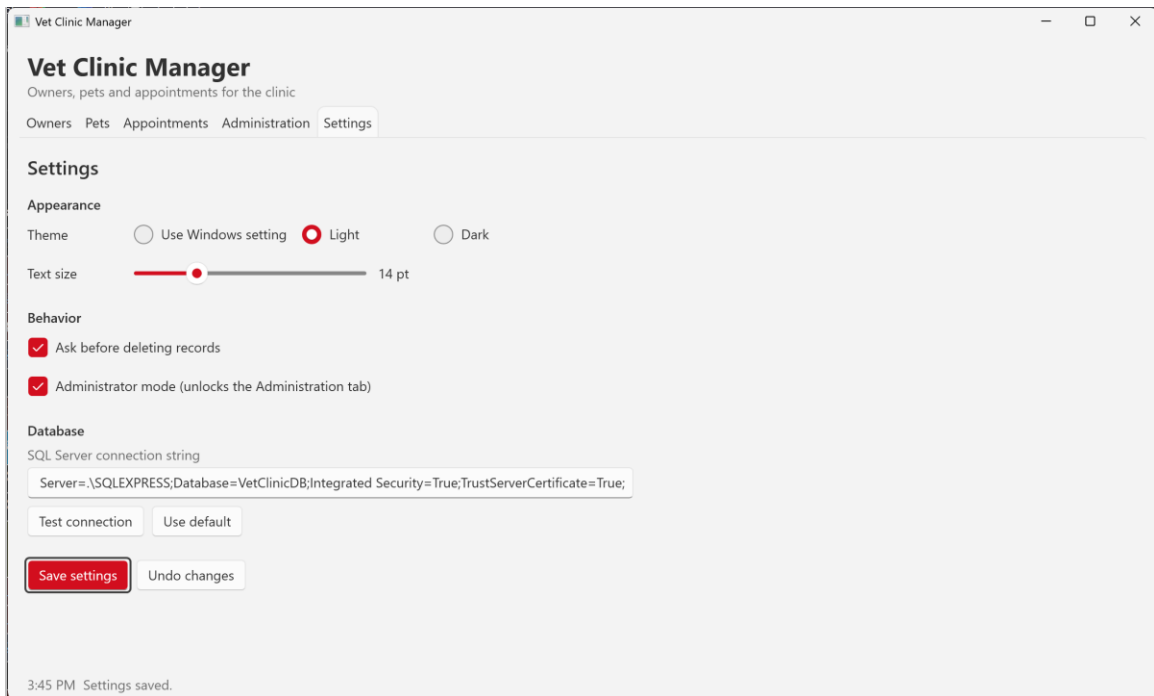


Figure 18. Settings: switching to the Light theme and enabling administrator mode, applied immediately.

Questions

1. Describe a challenge you overcame in this project

The hardest problem was getting the modern Fluent theme to apply to every control. I used WPF's new ThemeMode feature so the app could switch between light and dark mode, and I added app-wide implicit styles in App.xaml to give text boxes, buttons and data grids consistent margins. When I ran the app, the window was dark but the controls still had the old classic look, and the grid headers were white text on a white background.

I found the cause by removing pieces one at a time. First, assigning Application.Resources in XAML replaced the dictionary that ThemeMode had merged the Fluent styles into, so I merged Fluent.xaml explicitly. Second, my styles used BasedOn="{StaticResource {x:Type TextBox}}", which was resolved to the classic style instead of the Fluent one. Once I removed those styles the Fluent look came back, so I moved them into code (App.AddControlStyles), where FindResource returns the real Fluent base style. After that, light and dark switching worked at runtime from the Settings page.

2. Describe a technique you learned in this project

I learned how to write a safe, flexible search with ADO.NET. The Owners search lets the user pick a column (or any column) and type part of a value. The repository chooses the WHERE clause from a fixed list keyed by an enum, so the SQL text never contains user input, and the search term is passed as a parameter: @Pattern = '%' + term + '%'. I also learned that the characters %, _ and [are wildcards inside LIKE, so the code escapes them (for example, % becomes [%]) so a user's text is matched literally.

A second technique was letting the database enforce the rules. Pets and Appointments use ON DELETE CASCADE, so deleting an owner cleanly removes their pets and appointments, while the foreign key to Veterinarians has no cascade, so a veterinarian with history cannot be deleted. The data layer catches SQL error 547 and turns it into a friendly message instead of a crash.

3. What more would you want to refine to make it ready to sell, ship, and deploy?

- Real authentication and roles. Administrator mode is currently a checkbox; a shipping product needs sign-in (for example Windows accounts or Entra ID) with the admin role checked on the server.
- MVVM and tests. Moving the code-behind logic into view models would allow unit tests for validation and repository tests against a test database.
- Async database calls and paging, so the UI stays responsive with thousands of records.
- Packaging and code signing. While testing, Windows Smart App Control blocked one of my unsigned builds. A signed MSIX installer would avoid that and support automatic updates.
- Deployment of the database: a migration tool (such as DbUp or EF Core migrations), backups, and a central server instead of a local SQL Express instance.
- Logging and error reporting, plus accessibility and localization (more languages).

- More features for a real clinic: search on the Pets and Appointments tabs, a calendar view, double-booking checks, and invoices.

4. Which technology did you choose (WPF, WinForms, WinUI) and why?

I chose WPF. It is mature, well documented and runs on .NET 10 without extra packaging, so the project builds and runs with a plain dotnet build and no Windows App SDK setup, which WinUI 3 requires. Compared with WinForms, WPF's XAML layout (Grid, DockPanel), data binding and DataGrid made the list-and-detail screens quick to build, and they resize cleanly. WPF also includes the Fluent theme, which gives the app a modern Windows 11 look with light and dark modes. That made a theme setting easy to add to the Settings page.